

N 11.29

Дано:

Решение:

$$T = 373 \text{ K}$$

$$\Delta T = 1 \text{ K}$$

$$\text{Найти: } \frac{\Delta V}{V}$$

$$1) \frac{dP}{dT} = \frac{S_2 - S_1}{V_2 - V_1} \quad - \text{уравнение Клапейрона-Клаузиуса}$$

2) При конденсации газа пара:

$$\lambda \Delta m = m \Delta T \cdot \left(C_p - \frac{\lambda}{T} \right) \Rightarrow \frac{\Delta m}{m} = \left(\frac{C_p T}{\lambda} - 1 \right) \frac{\Delta T}{T} \approx 9,17\%$$

$$\delta Q = dH - V dp = C_p dT - V dp = C dT; C = C_p - \frac{\lambda}{T}$$

N 11.74

Дано:

Решение:

$$S_x = \frac{RT}{\Theta}$$

$$S_{TB} = 0,7R$$

$$T_1 = 0,25$$

$$P_1 = 29 \text{ атм}$$

$$T = 0,1 \text{ K}$$

$$\Delta V = 1,25 \text{ см}^3$$

$$\text{Найти: } P_2$$

$$\frac{dP}{dT} = \frac{S_2 - S_1}{V_2 - V_1} = \frac{S_T - S_x}{\Delta V} = \frac{-0,7R + \frac{RT}{\Theta}}{\Delta V}$$

$$P - P_1 = \left(0,7RT \Big|_{T_1}^T + \frac{RT^2}{2\Theta} \Big|_{T_1}^T \right) \frac{1}{\Delta V} \Rightarrow$$

$$\Rightarrow P = 3,22 \cdot 10^6 \text{ Па} \approx 32,2 \text{ атм}$$

N 12.49

Дано:

Решение:

$$T = 293 \text{ K}$$

$$r_1 = 10^{-7} \text{ м}$$

$$r_2 = 10^{-9} \text{ м}$$

$$\sigma = 72,7 \cdot 10^{-3} \frac{\text{Н}}{\text{м}}$$

$$v_x = 1,002 \frac{\text{см}^3}{2}$$

$$P_0 = 17,5 \text{ атм}$$

$$P_{\text{н.п.}} = P_0 + \frac{P_{\text{н.п.}}}{P^*} \leq K; P_{\text{н.п.}} = \frac{P_{\text{н.п.}}}{RT}$$

$$dA = \frac{P_{\text{н.п.}}}{RT} dK$$

$$K = \frac{P_{\text{н.п.}}}{RT}$$

Рн.п.

$$\int_{P_0} \frac{dP}{P} = \int_0^r \frac{\mu \sigma dK}{RT P^*}$$

$$P = P_0 \exp\left(\frac{2\mu \sigma}{RT P^* r}\right)$$

№ 2.49

Дано:

Решение:

$$T = 293 K$$

$$r_1 = 10^{-7} \text{ м}$$

$$r_2 = 10^{-8} \text{ м}$$

$$\sigma = 72,7 \frac{\text{Дж}}{\text{см}^2}$$

$$\nu_x = 1,002 \frac{\text{см}^3}{\text{г}}$$

$$P_0 = 17,5 \text{ мм. рт. ст.}$$

№ 2.15

Дано:

$$V_0 = 5 \text{ моль}$$

$$\gamma = \frac{4}{3}$$

$$T_0 = 300 K$$

$$\Delta V = 1$$

Найти: T

$\Delta U =$ кол-во энергии, уносимой

поточкой газа

$$C_p T dV = C_v V dT - C_v (V - dV)(T - dT)$$

$$C_p T dV = C_v \cancel{V} T - C_v \cancel{V} T + C_v T dV + C_v V dT + o.c.$$

$$R T dV = C_v V dT$$

$$\frac{dV}{V} = \frac{dT}{T} \frac{C_v}{R} \Rightarrow T = T_0 \cdot \left(\frac{V_0 - \Delta V}{V}\right)^{\frac{R}{C_v} \gamma - 1} = 278 (K)$$

N6. 39

Дано:

$$T_1 = 680 \text{ K}$$

$$T_2 = 250 \text{ K}$$

$\gamma = 1.4$ моль

Найти: ΔS

Решение:

$$1) S(T, V) \Rightarrow dS = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV =$$

$$= C_V \frac{dT}{T} + \frac{R dV}{V-b}$$

$$dF = -S dT - p dV; \left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial p}{\partial T} \right)_V$$

$$p = \frac{RT}{V-b} - \frac{a}{V^2} \Rightarrow \left(\frac{\partial p}{\partial T} \right)_V = \frac{R}{V-b}$$

$$\Rightarrow S = C_V \ln \frac{T}{T_0} + R \ln \frac{V-b}{V_0-b}$$

$$2) \left(p + \frac{a}{V^2} \right) (V-b)^3 = \text{const}; \left(p + \frac{a}{V^2} \right) (V-b) = RT \Rightarrow$$

$$\Rightarrow \left(p + \frac{a}{V^2} \right) = \frac{RT}{V-b} \Rightarrow (V-b)^2 \cdot T = \text{const}$$

$$3) \frac{(V-b)^2}{(V_0-b)^2} = \frac{T_0}{T} \Rightarrow$$

$$\Rightarrow \Delta S = C_V \ln \frac{250}{680} + \frac{1}{2} R \ln \frac{680}{250} = 2R \ln \frac{250}{680} \approx$$

$$C = \frac{D^*}{\dots}$$

$\lambda = 16,6$ мкм.к.

N 6.17

Дано:

$$\rho_x = 1,9 \rho_{np}$$

$$V_x = \frac{V_0}{2}$$

$$T_{кр} = 467 K$$

Найти: T

Решение:

$$1) \left(\pi + \frac{3}{\varphi^2} \right) \left(\varphi - \frac{1}{3} \right) = \frac{8}{3} \kappa$$

$$2) \rho_{np} \cdot V_0 = 1,9 \rho_{np} \cdot \frac{V_0}{2} + \rho_n \cdot \frac{V_0}{2}$$

$$2 \rho_{np} = 1,9 \rho_{np} + \rho_n \Rightarrow \rho_n = 0,1 \rho_{np}$$

3) Худ-то и пар газы удовлетворяют (1):

$$\begin{cases} \left(\pi + \frac{3}{\varphi_1^2} \right) \left(\varphi_1 - \frac{1}{3} \right) = \frac{8}{3} \kappa & ; \varphi_1 = \frac{V_{\pi}}{V_{\kappa}} = \frac{\rho_{\kappa}}{\rho_{\pi}} = 10 \\ \left(\pi + \frac{3}{\varphi_2^2} \right) \left(\varphi_2 - \frac{1}{3} \right) = \frac{8}{3} \kappa & \varphi_2 = \frac{V_x}{V_{\kappa}} = \frac{\rho_{\kappa}}{\rho_n} = \frac{10}{19} \end{cases}$$

$$4) \begin{cases} \left(\pi + 0,03 \right) \left(10 - \frac{1}{3} \right) = \frac{8}{3} \kappa \\ \left(\pi + \frac{3 \cdot 19^2}{100} \right) \left(\frac{10}{19} - \frac{1}{3} \right) = \frac{8}{3} \kappa \end{cases} \quad \left| \begin{array}{l} \pi = 0,19; \kappa \approx 0,8 \end{array} \right.$$

$$\Rightarrow T = 0,8 T_{кр} = 374 (K)$$

Ответ: $T = 374 K$.